

# NAMIT RATHOD

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## SUMMARY

AI Engineer with 3+ years building agentic AI systems and LLM pipelines across healthcare tech. Hands-on with **MCP, LangGraph, Azure OpenAI, Claude, NeMo Guardrails, GraphRAG, and FHIR** - shipped clinical AI products cutting manual workflows by 60-87%. M.S. Computer Science, University of Houston (GPA 3.52).

## EDUCATION

University of Houston-Clear Lake - M.S. Computer Science | GPA: 3.52

Aug 2024 - May 2026

## WORK EXPERIENCE

**AI Engineer Intern** | ScriptChain Health | Remote

Jan 2026 - May 2026

- **Deployed LLM care plan microservice** via **FastAPI** + **LangGraph**, integrating FHIR EHR and food/exercise DB as Claude-callable **MCP** tools; cut LLM context prep by **60%**
- **Built GraphRAG pipeline** over patient EHR records using **PyTorch** embeddings, enabling graph-aware clinical reasoning that improved care plan relevance vs. flat vector retrieval
- **Enforced 100% clinical guideline compliance** by integrating **NVIDIA NeMo Guardrails** into multi-agent orchestration, aligning all AI outputs with AHA and U.S. Dietary Guidelines
- **Reduced EHR payload 80%** by designing **GraphQL** abstraction over Epic/Cerner FHIR APIs, enabling real-time patient data delivery to AI inference pipelines
- **Cut deployment time 82% and release errors 70%** via **Docker** + **CI/CD** with automated **Pytest** suites and PR validation gates

**Software Engineer Intern** | Nasafitness LLC | Houston, TX

Aug 2025 - Dec 2025

- **Eliminated manual clinical documentation** for 100+ sessions by engineering **Azure OpenAI GPT-4o** + **.NET 8** pipeline synthesizing Telehealth transcripts into structured wellness care plans
- **Improved inference accuracy and cut token costs 80%** via context pruning algorithm; guaranteed 100% deterministic JSON via **Semantic Kernel** orchestration
- **Cut referral processing 87% (15 min to 2 min)** via **RingCentral API**; maintained 99.9% uptime across 6 microservices with full **CI/CD** and Azure observability

**Software Engineer** | Cognizant | Bengaluru, India

Jan2022 - Jul 2024

- **Reduced query latency 35% across 1M+ records** by redesigning **PostgreSQL** schemas with indexing; scaled **Java Spring Boot** microservices to 1,000+ concurrent users
- **Achieved 99.9% build success rate** via **Jenkins** + **GitLab CI**; cut MTTD 20% with **Prometheus** + **Grafana** observability

## PROJECTS

**Fair-KV Cache** | Python, FastAPI, Docker, Prometheus, OpenAI API, YAML

- **Eliminated KV cache monopolization in multi-tenant LLM serving** via per-tenant quotas, oldest-first eviction, and elastic borrowing (1.5x idle capacity); drove victim P95 latency to near-zero under adversarial burst, validated with Jain's fairness index across LRU, strict-quota, and FairKV policies

**InboxSense: Semantic Email Intelligence** | Next.js 14, Gemini 2.5, ChromaDB, RAG, Prisma, OAuth 2.0

- **Reduced email noise by 70%** by engineering a **RAG pipeline** converting Gmail data into 3072-dim vectors via gemini-embedding-2-preview; built resilient background sync with exponential backoff and privacy-first Dockerized **ChromaDB** storage

**Real-Time Payment Settlement Engine** | Java 21, Spring Boot, Kafka, Redis, PostgreSQL, Gatling, Docker

- **Processed 1,000+ concurrent transactions at sub-100ms** via Java 21 Virtual Threads and Redis idempotency layer; validated 97.99% success rate under Gatling peak-load simulation with zero duplicate charges

## TECHNICAL SKILLS

**AI & LLM:** LangGraph, MCP, Azure OpenAI, Claude, GPT-4o, NeMo Guardrails, Semantic Kernel, RAG, GraphRAG, PyTorch, Embeddings, ChromaDB, Prompt Engineering, FHIR AI Pipelines

**Languages:** Python, Java, C#, Go, Rust, TypeScript, SQL

**Backend & APIs:** FastAPI, Spring Boot, .NET 8, GraphQL, REST, gRPC, Kafka, Celery, Redis, Microsoft Graph API

**Data:** PostgreSQL, SQL Server, Cosmos DB, DynamoDB, MongoDB, Apache Arrow, KV cache design, schema design

**Cloud & DevOps:** AWS (EC2, S3, ECS, Lambda, SES), Azure, Docker, Kubernetes, GitHub Actions, Jenkins, GitLab CI

**Testing & Observability:** Pytest, JUnit, Jest, Playwright, Prometheus, Grafana, Application Insights, Elasticsearch